Ministry of Higher Education and Scientific Research
Scientific Supervision and Scientific Evaluation Apparatus
Directorate of Quality Assurance and Academic Accreditation
Accreditation Department



Academic Program and Course Description Guide

2025

The Introduction

The educational program is a coordinated and organized package of courses that include procedures and experiences organized into study modules. The ,primary purpose of the program is to build and refine the skills of graduates making them qualified to meet the requirements of the labor market. It is reviewed and evaluated annually through internal or external audit .procedures and programs, such as the External Examiner Program

The academic program description provides a brief summary of the program's main features and courses, indicating the skills that students are expected to acquire based on the program's objectives. The importance of this description is evident in that it represents the cornerstone of program accreditation and is written by faculty members under the supervision of .academic committees in the academic departments

This guide, in its second edition, includes a description of the academic program after updating the vocabulary and paragraphs of the previous guide in light of the new developments and changes in the educational system in Iraq, which included a description of the academic program in its traditional form (annual, semester) in addition to adopting the description of the academic program circulated pursuant to the letter of the Department of Studies TM 2906/3 dated 5/3/2023 with regard to programs that adopt the .Bologna process as the basis for their work

In this context, we cannot but emphasize the importance of writing academic program and course descriptions to ensure the smooth running of the .educational process

Concepts and terms:

:Academic program description The academic program description provides a concise summary of the program's vision, mission, and objectives, including a precise description of the targeted learning outcomes according to specific .learning strategies

Course Description: Provides a concise summary of the course's key features and the learning outcomes expected of the student, demonstrating whether the student has made the most of the available learning opportunities. It is .derived from the program description

Program Vision: An ambitious vision for the future of the academic program .to be an advanced, inspiring, motivating, realistic, and applicable program

Program mission: It briefly explains the objectives and activities required to achieve them, and also identifies the program's development paths and .directions

Program objectives: These are statements that describe what the academic program intends to achieve within a specific time period and are measurable .and observable

Curriculum structure: All courses/subjects included in the academic program ,according to the approved learning system (semester-year, Bologna track whether required by (ministry, university, college, or scientific department) .with the number of academic units

Learning outcomes: A consistent set of knowledge, skills, and values acquired by a student after successfully completing the academic program. Learning outcomes for each course must be defined in a way that achieves the .program's objectives

Teaching and learning strategies: These are the strategies used by a faculty member to develop student teaching and learning. They are plans followed

to achieve learning objectives, i.e. they describe all classroom and .extracurricular activities to achieve the learning outcomes of the program

Academic Program Description Form

...University name: University of...Basra

.....College/Institute: College of .....Engineering

....Academic Department: Department of .....Mechatronics Engineering

Name of academic or professional program: Bachelor of Mechatronics

.....Engineering

....Final Degree Name: Bachelor of Mechatronics Engineering

track Semester / Bologna

.....Description preparation date: 2025

....Date of filling out the file: 11/9/2025

:the signature 

.Name of Scientific Assistant: Prof. Dr Munir Abdel Jalil Ismail

:the date

:the signature 

.Department Head Name: Asst. Prof. Dr Alaa Falah Abdul Hassan

Date : 12/9/2025

Check the file before

Quality Assurance and University Performance Division

Name of the Director of the Quality Assurance and University Performance Marzouk Division : Asst. Prof. Dr. Ali Kamel

the date

the signature 



Dean's approval

Program vision .1

- Upgrading the department's infrastructure, education, and digital and smart skills to support the university's drive towards becoming a fourth-generation university.
- Raising academic, scientific and administrative work to a level that leads to obtaining international accreditation.
- Striving to meet all requirements for opening postgraduate studies in the department in the future.
- Striving to involve the department in serving the community by contributing to industrial, construction and service projects that fall within the department's specialization in service of the nation and society.

Program message .2

- Developing students' learning skills to ensure they achieve outstanding academic standards in accordance with educational outcomes.
- Meeting the quality requirements of in-person and online education, with the aim of achieving blended learning, which will lead to diverse, modern, and effective learning outcomes.

Creating an interactive, scientific and practical academic creativity environment and supporting students' classroom and extracurricular activities.

Program objectives .3

- The department aims to introduce new and advanced concepts in the field of industry and manufacturing, and to employ programming to modernize, and develop various industrial systems to enhance their accuracy, performance, and speed.

- ,Employing all types of artificial intelligence in the development modernization, and upgrading of traditional mechanical and electrical .systems
- Providing graduates with the necessary experience to integrate modern automated control systems with traditional pneumatic, hydraulic, and .mechanical systems
- Providing graduates with the necessary expertise in the areas of linking medical devices and tools with programmed electronic systems to support the health sector, particularly in the field of medical robotics and the .manufacture of smart prosthetic limbs
- Qualifying and training students to achieve a level that makes them desirable in the job market and providing them with engineering and administrative experience, reinforced by professional ethics, to develop .the public and private industrial sector and raise it to prestigious levels

Program accreditation .4

Other external influences .5

Program structure .6

*comments	percentage	Study unit	Number of courses	Program structure
Basic	**10%**	European 12 units	**6**	Institutional requirements
Basic	**27%**	European 59 units	**15**	College requirements

Basic	**63%**	European 169 units	**35**	Department requirements
essential			existing	Summer training
				Other

.Notes may include whether the course is core or elective*

Program Description .7				
Credit hours		Course name	Course code	Year/Level
practical	theoretical			
	3	Mathematics**I**	**MCT111**	First/First Level/First Semester
2	**3**	Electrical Circuits 1	**MCT112**	First/First Level/First Semester
2	**2**	Computer Principles	**MCT113**	First/First Level/First Semester
	2	chemistry	**MCT114**	First/First Level/First Semester
	3	Engineering Mechanics (Statics)	**MCT115**	First/First Level/First Semester
	2	Arabic language 1	**MCT116**	First/First Level/First Semester
4	**--**	Computer engineering drawing	**MCT117**	First/First Level/First Semester
	3	Mathematics**II**	**MCT121**	First/Second Level/Second Semester
2	**3**	Electrical Circuits 2	**MCT122**	First/Second Level/Second Semester
2	**3**	Digital Systems Design	**MCT123**	First/Second Level/Second Semester
	2	physics	**MCT124**	First/Second Level/Second Semester
	3	Engineering Mechanics (Dynamics)	**MCT125**	First/Second Level/Second Semester
	2	English 1	**MCT126**	First/Second Level/Second Semester

	2	Human rights and democracy and freedom	**MCT127**	First/Second Level/Second Semester
	3	Mathematics**III**	**MCT211**	Second/Second Level/Third Term
2	**3**	material resistance	**MCT212**	Second/Second Level/Third Term
	2	Metallurgical Engineering	**MCT213**	Second/Second Level/Third Term
2	**2**	Computing and Artificial Intelligence	**MCT214**	Second/Second Level/Third Term
	2	Electrical machines (DC)	**MCT215**	Second/Second Level/Third Term
	2	English 2	**MCT216**	Second/Second Level/Third Term
	2	thermodynamics	**MCT217**	Second/Second Level/Third Term
	3	differential equations	**MCT221**	Second/Second Level/Fourth Semester
2	**2**	electrical machines alternating) (current	**MCT222**	Second/Second Level/Fourth Semester
2	**3**	Object-oriented programming	**MCT223**	Second/Second Level/Fourth Semester
	3	Machine theory	**MCT224**	Second/Second Level/Fourth Semester
2	**3**	electronics	**MCT225**	Second/Second Level/Fourth Semester
	2	Baath Party crimes	**MCT226**	Second/Second Level/Fourth Semester
	2	Arabic 2	**MCT227**	Second/Second Level/Fourth Semester
	3	Linear algebra	**MCT311**	/Third/ Third Level Fifth Term
2	**3**	mechanical vibrations	**MCT312**	/Third/ Third Level Fifth Term
	2	Fluid Mechanics and Systems	**MCT313**	/Third/ Third Level Fifth Term
	3	Control Systems 1	**MCT314**	/Third/ Third Level Fifth Term
2	**3**	Computer organization	**MCT315**	/Third/ Third Level Fifth Term
	3	Power electronics	**MCT316**	/Third/ Third Level Fifth Term

	2	heat transfer	**MCT317**	/Third/ Third Level Fifth Term
	3	Numerical analysis and statistics	**MCT321**	/Third/ Third Level Sixth Chapter
	2	Engineering Economics	**MCT322**	/Third/ Third Level Sixth Chapter
	2	Sensors and microcontrollers	**MCT323**	/Third/ Third Level Sixth Chapter
2	**3**	Control Systems 2	**MCT324**	/Third/ Third Level Sixth Chapter
2	**2**	Automation systems	**MCT325**	/Third/ Third Level Sixth Chapter
	2	Digital signal processing	**MCT326**	/Third/ Third Level Sixth Chapter
	2	renewable energy	**MCT327**	/Third/ Third Level Sixth Chapter
2	**3**	Embedded systems	**MCT411**	/ Fourth / Fourth Level Seventh Chapter
	3	Mechanical Design 1	**MCT412**	/ Fourth / Fourth Level Seventh Chapter
	3	artificial intelligence	**MCT413**	/ Fourth / Fourth Level Seventh Chapter
3	**2**	Engineering project	**MCT4P**	/ Fourth / Fourth Level Seventh Chapter
	2	project management	**MCT414**	/ Fourth / Fourth Level Seventh Chapter
2	**3**	Mechatronics - Systems Design UAVs	**MCT415**	/ Fourth / Fourth Level Seventh Chapter
	2	industrial communication networks	**MCT416**	/ Fourth / Fourth Level Seventh Chapter
	3	Biomechatronics	**MCT421**	/ Fourth / Fourth Level Eighth Chapter
2	**3**	Mechanical Design 2	**MCT422**	/ Fourth / Fourth Level Eighth Chapter
	2	Industrial Engineering	**MCT423**	/ Fourth / Fourth Level Eighth Chapter
	3	Robotic systems	**MCT424**	/ Fourth / Fourth Level Eighth Chapter
3	**2**	Engineering project	**MCT4P**	/ Fourth / Fourth Level Eighth Chapter
	2	Professional ethics	**MCT425**	/ Fourth / Fourth Level Eighth Chapter
2	**3**	Image processing and computer vision	**MCT426**	/ Fourth / Fourth Level Eighth Chapter

Expected learning outcomes of the program .8	
knowledge	
Learning Outcomes Statement 1	Learning Outcomes 1
Skills	
Learning Outcomes Statement 2	Learning Outcomes 2
Learning Outcomes Statement 3	Learning Outcomes 3
values	
Learning Outcomes Statement 4	Learning Outcomes 4
Learning Outcomes Statement 5	Learning Outcomes 5

Teaching and learning strategies .9

1- Explanation and clarification through lectures and increasing understanding through .laboratories

.2 ,How to display scientific materials using projectors: data show, smart boards, plasma screens .tablets

3 .Self-learning through homework

4 .Weekly discussion of the subject and solving some additional questions on the subject

Evaluation methods .10

.1 Short exams (coz)

2 .Homework .

3 Midterm and final exams for theoretical subjects . .

.4 Interaction within the lecture and student attendance .

.Absence, attendance and general commitment .5

Faculty .11

Faculty members

Faculty preparation		Special requirements skills (if any)/		Specialization		Academic rank
lecturer	angel			private	general	
	permanent			Control and computers	Electrical Engineering	Dr. Ali Ahmed Abdel

	permanent			Robotic systems	Electrical Engineering	Dr. Alaa Falah Abdul Hassan
	permanent			Electricity and computers	Electrical Engineering	.Dr Muhannad Hamed Khalaf
	permanent			Heat	Mechanical Engineering	Dr. Sanaa Jafar Yassin
	permanent			Heat	Mechanical Engineering	D. Amani Jalil Majeed
	permanent			Control and computers	Electrical Engineering	Dr. Samia Daoud Shaker
	permanent			Heat	Mechanical Engineering	M.M. Firas Matar Khalaf Matar
	permanent			Control and computers	Electrical Engineering	M.M. Youssef Abdel Wahab Khairallah
	permanent			internationa l law	law	M.M. Ahmed Shatti Ghadhban
	permanent			mechanizati on	agriculture	M.M. Hussein Salman Majeed
	permanent			Power and machinery	Electrical Engineering	M.M. Ilaf Jalil Majeed
	permanent			Heat	Mechanical Engineering	.millimeter Abbas Fadel Sultan
	permanent			Materials Technologies	Petrochemical Engineering Technology	.M.M Mohammed Jassim Mohammed Radhi
lecturer				Calculators	Computer Engineering	M.M. Rafid Mohammed Khalifa

Professional development

Orientation of new faculty members

The department is concerned with guiding new faculty members, by introducing them to the .department's mission and educational objectives
Providing them with the necessary guidance on effective teaching strategies, modern .assessment methods, and mechanisms for integrating technology into the educational process
Orientation sessions and training workshops are also organized to help new teachers adapt to the

The academic work environment, clarifying their teaching, research, and administrative responsibilities, and enhancing communication between new faculty members and their more .experienced colleagues through academic supervision and career guidance programs

Professional development for faculty members

.1 .Training workshops on modern teaching methods and active learning techniques
.2 .Development courses in digital technologies to support the educational process
.3 .Using e-learning tools to support the educational process
.4 Academic supervision and guidance programs to exchange experiences between new faculty .members and experienced faculty members
.5 Encouraging scientific research, publishing in reputable journals, and participating in scientific .conferences and seminars

Acceptance Criteria .12

1 Average: not less than 90% .
.2 Age: not more than 25 years
3 Number: about 60 students per year .

The most important sources of information about the program .13

1. Websites of official Iraqi academic institutions
2. Workshops of the Ministry of Higher Education and Scientific Research
3. **ABET** Accreditation Program

Program Development Plan .14

The Department of Mechatronics Engineering seeks to develop its academic program :according to the principle of continuous improvement, through

1. Updating curricula to include modern topics such as renewable energy, artificial .intelligence, automation, and smart systems
2. Improving teaching and assessment methods through e-learning and continuous training .for teaching and technical staff
3. Supporting industrial scientific research through cooperation with academic and .industrial institutions
4. Updating infrastructure and laboratories
5. .Enhancing student and community services

The boxes corresponding to the individual learning outcomes of the programme being .assessed have been checked																			
Required learning outcomes of the program																			
General and) transferable skills (other skills Related to employability and personal development				Affective and value-based goals				Program skill objectives				Cognitive objectives				Basic or optional	Course name	Course code	Year Leve/ l
4 d	3 d	2 d	1 d	4 C	3 C	2 C	1 c	4 b	3 b	2 b	1 b	4 A	3 A	2 A	1 A				
✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	essential	Mathematics 1	MCT111	First
✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	essential	Electrical Circuits 1	MCT112	
✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	essential	Computer 1	MCT113	
✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	essential	Chemistry	MCT114	
✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	essential	Engineering Mechanics (Statics)	MCT115	
✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	essential	Arabic language 1	MCT116	
✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	essential	Computer engineering drawing	MCT117	
✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	essential	Mathematics 2	MCT121	

✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	essential	Electrical Circuits 2	**MCT122**	First
✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	essential	Digital Systems Design	**MCT123**	
✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	essential	physics	**MCT124**	
✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	essential	Engineering Mechanics (Dynamics)	**MCT125**	
✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	essential	English 1	**MCT126**	
✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	essential	Human rights and democracy and freedom	**MCT127**	

✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	essential	Mathematics**III**	**MCT211**	
✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	essential	Strength of material	**MCT212**	
✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	essential	Metallurgical Engineering	**MCT213**	
✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	essential	Computing and Artificial Intelligence	**MCT214**	
✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	essential	Electrical machines (DC)	**MCT215**	Second
✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	essential	English 2	**MCT216**	
✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	essential	Thermodynamic	**MCT217**	
✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	essential	Differential equations	**MCT221**	
✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	essential	Electric machines alternating) (current	**MCT222**	
✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	essential	Object oriented programing	**MCT223**	
✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	essential	Machine theory	**MCT224**	
✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	essential	electronics	**MCT225**	second
✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	essential	Baath Party crimes	**MCT226**	
✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	essential	Arabic 2	**MCT227**	

✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	essential	Linear algebra	**MCT311**	Third
✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	essential	mechanical vibrations	**MCT312**	
✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	essential	Fluid Mechanics and Systems	**MCT313**	
✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	essential	Control Systems 1	**MCT314**	
✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	essential	Computer organization	**MCT315**	
✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	essential	Power electronics	**MCT316**	
✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	essential	heat transfer	**MCT317**	
✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	essential	Numerical analysis and statistics	**MCT321**	Third
✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	essential	Engineering economics	**MCT322**	
✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	essential	Sensors and microcontrollers	**MCT323**	
✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	essential	Control Systems 2	**MCT324**	
✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	essential	Automation systems	**MCT325**	
✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	essential	Digital signal processing	**MCT326**	
✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	essential	Renewable energy	**MCT327**	

✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	essential	Embedded systems	**MCT411**	Fourth
✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	essential	Mechanical Design 1	**MCT412**	
✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	essential	Artificial intelligence	**MCT413**	
✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	essential	Engineering project	**MCT4P**	
✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	essential	project management	**MCT414**	
✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	essential	Mechatronics - Systems Design UAVs	**MCT415**	
✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	essential	industrial communication networks	MCT416	
✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	essential			
✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	essential	Biomechatronics	**MCT421**	Fourth
✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	essential	Mechanical Design 2	**MCT422**	
✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	essential	industrial engineering	**MCT423**	
✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	essential	Robotic systems	**MCT424**	
✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	essential	Engineering project	**MCT4P**	
✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	essential	Professional ethics	**MCT425**	
✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	essential	Image processing and computer vision	**MCT426**	
✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	essential			

Course Description Form

Course Name: Mathematics 1 .1
:Course code .2**MCT** 111
Semester/Year: First Semester - First Stage .3
Date this description was prepared: 2025 .4
.Available attendance forms: in-class attendance, online attendance .5
Number of study hours (total) / Number of units (total): 150 hours / 6 .6**ECTS**
:Name of the course supervisor (if more than one name is mentioned) .7
:Name: M.M. Youssef Abdel Wahab Khairallah Email

Course objectives .8	
Learning definite and indefinite integration, differentiation, graphing functions, and finding limits.	Course objectives
Attending lectures, viewing video lectures, solving homework assignments, and solving some questions from the prescribed book.	Strategy

Course environment .10

Evaluation method	Learning method	Name of unit or topic	Required learning outcomes	watches	week
duties, jugs	,Attendance duties	Prerequisites for calculus	Learn the basics of mathematics	3	1
duties, jugs	,Attendance duties	Functions	Learn types of functions	3	2
duties, jugs	,Attendance duties	Graph of functions	Graphing functions	3	3
duties, jugs	,Attendance duties	Graph of functions	Graphing functions	3	4

duties, jugs	,Attendance duties	Limits	Goals	3	5
duties, jugs	,Attendance duties	Continuity	Continuity	3	6
duties, jugs	,Attendance duties	Derivatives	derivatives	3	7
duties, jugs	,Attendance duties	Derivatives	derivatives	3	8
duties, jugs	,Attendance duties	Implicit differentiation	implicit derivation	3	9
duties, jugs	,Attendance duties	Application of derivatives	Derivative applications	3	10
duties, jugs	,Attendance duties	Applications of derivatives	Derivative applications	3	11
duties, jugs	,Attendance duties	Integration	Indefinite integrals	3	12
duties, jugs	,Attendance duties	integration	Finite integrals	3	13
duties, jugs	,Attendance duties	Applications	Integration applications	3	14
duties, jugs	,Attendance duties	Discussion	General discussion	3	15

Course Evaluation .11	
Jugs, homework, college assignments, reports	
Learning and teaching resources .12	
Thomas' Calculus Early Transcendentals	Required textbooks (methodology if (available
	Main References (Sources)
	Recommended supporting books and references (scientific journals, reports...)
A Textbook of ENGINEERING MATHEMATICS-I	Electronic references, websites

Course Description Form

Course Name: Electrical Circuits 1 .1	
:Course code .2**MCT** 112	
Semester/Year: First Semester - First Stage .3	
Date this description was prepared: 2025 .4	
.Available attendance forms: in-class attendance, online attendance .5	
Number of study hours (total) / Number of units (total): 150 hours / 6 .6**ECTS**	
:Name of the course supervisor (if more than one name is mentioned) .7	
:Name: Asst . Prof. Dr. Alaa Falah Abdul Hassan Email	
Course objectives .8	
1. Understanding basic circuit components, such as resistors, capacitors, and inductors, and their properties. 2. Familiarizing with various types of circuits, such as series, parallel, and combination circuits. 3. Analyzing DC circuits using different analysis techniques. 4. Analyzing AC circuits using complex impedance and phasor notation. 5. Understanding the behavior of circuits with reactive components. 6. Understanding the concept of power and energy in circuits. Developing practical skills in designing and building basic electrical circuits.	اهداف المادة الدراسية
The main strategy that will be adopted in delivering this module is to encourage students' participation in the exercises, while at the same time refining and expanding their critical thinking skills. This will be achieved through classes, interactive tutorials and by considering type of simple experiments involving some sampling activities that are interesting to the students.	Strategy
Course environment .10	

ation method	rning method	me of unit or topic	equired learning outcomes	atches	veek
duties, jugs	,Attendance duties	Introduction		3	1
duties, jugs	,Attendance duties	Basic network elements		3	2
duties, jugs	,Attendance duties	Resistance, resistivity, ohm's law		3	3
duties, jugs	,Attendance duties	Kirchhoff's laws		3	4
duties, jugs	,Attendance duties	Nodal analysis		3	5
duties, jugs	,Attendance duties	Mesh analysis		3	6
duties, jugs	,Attendance duties	Superposition theorem		3	7
duties, jugs	,Attendance duties	Thevenin's analysis		3	8
duties, jugs	,Attendance duties	Norton analysis		3	9
duties, jugs	,Attendance duties	Source transformation		3	10
duties, jugs	,Attendance duties	Maximum power transfer		3	11
duties, jugs	,Attendance duties	Star Connection		3	12
duties, jugs	,Attendance duties	Delta connection		3	13
duties, jugs	,Attendance duties	Star/delta transformation		3	14
duties, jugs	,Attendance duties	Transient response		3	15

Course Evaluation .11

Jugs, homework, college assignments, reports

Learning and teaching resources .12

Fundamentals of Electrical Circuits, CK Alexander and MNO Sadiku, McGraw-Hill Education	Required textbooks (methodology if (available
	Main References (Sources)
	Recommended supporting books and ,references (scientific journals (...reports
Introductory Circuit Analysis, R. Boylestad , Pearson	Electronic references, websites

Course Description Form

Course Name: Computer Principles .1
:Course code .2**MCT** 113
Semester/Year: First Semester - First Stage .3
Date this description was prepared: 2025 .4
.Available attendance forms: in-class attendance, online attendance .5
Number of study hours (total) / Number of units (total): 75 hours / 3 .6**ECTS**
:Name of the course supervisor (if more than one name is mentioned) .7
:Name: Asst. Dr. Samia Daoud Shaker Email

Course objectives .8	
1. Utilize the computer for fundamental tasks. 2. Identify and discuss the hardware components of the computer system. 3. Creating documents using a word processor and creating presentations. 4. An introduction to the internet and communications.	Course objectives
The main strategy that will be adopted in delivering this module is to encourage students' participation in the exercises, while at the same time refining and expanding their critical thinking skills. This will be achieved through classes, interactive tutorials and by considering type of simple experiments involving some sampling activities that are interesting to the students.	Strategy

Course environment .10

Evaluation method	arning method	Name of unit or topic	Required learning outcomes	vatches	week
duties, jugs	,Attendance duties	Introduction		2	1
duties, jugs	,Attendance duties	Computer components		2	2

duties, jugs	,Attendance duties	Computer components		2	3
duties, jugs	,Attendance duties	OS and GUI		2	4
duties, jugs	,Attendance duties	OS and GUI		2	5
duties, jugs	,Attendance duties	Word processing		2	6
duties, jugs	,Attendance duties	Word processing		2	7
duties, jugs	,Attendance duties	Spread sheets		2	8
duties, jugs	,Attendance duties	Presentation software		2	9
duties, jugs	,Attendance duties	Internet and web browser		2	10
duties, jugs	,Attendance duties	Internet and web browser		2	11
duties, jugs	,Attendance duties	Communication and emails		2	12
duties, jugs	,Attendance duties	Communication and emails		2	13
duties, jugs	,Attendance duties	Computer troubleshooting		2	14
duties, jugs	,Attendance duties	Computer troubleshooting		2	15

Course Evaluation .11	
Jugs, homework, college assignments, reports	
Learning and teaching resources .12	
Graham Brown, David Watson, ''Cambridge IGCSE information and communication technology'', 3rd Edition (2020)	Required textbooks (methodology if (available
	Main References (Sources)
	Recommended supporting books and ,references (scientific journals (...reports
Alan Evans, Kendall Martin, Mary Anne Poatsy, ''Technology in action complete'', 16th Edition (2020)	Electronic references, websites

Course Description Form

Course name: Chemistry .1	
:Course code .2**MCT** 114	
Semester/Year: First Semester - First Stage .3	
Date this description was prepared: 2025 .4	
.Available attendance forms: in-class attendance, online attendance .5	
Number of study hours (total) / Number of units (total): 75 hours / 3 .6**ECTS**	
:Name of the course supervisor (if more than one name is mentioned) .7	
:Name: M.M. Mohammed Jassim Mohammed Radhi Email	
Course objectives .8	
1. Describe the fundamental issues of chemical reactions, equilibrium and kinetics. Study the considerations of industrial chemistry such as reaction evaluation and types of industrial reactors. Depict the chemistry of gas and petroleum. Elaborate on the Chemistry of ethylene and propylene and treat the C4 and C5 olefins. Discuss the Chemistry of the benzene, toluene, and the xylenes. 2. Study some topics in bioengineering that are related to computer engineering with emphasis on neural networks.	Course objectives
1. Class lectures. 2. Tutoring. 3. Homework. 4. quizzes 5. Mid-term and final exams	Strategy
Course environment .10	

Evaluation method	Learning method	Name of unit or topic	Required learning outcomes	watches	week
duties, jugs	,Attendance duties	Chemical reactions		2	1
duties, jugs	,Attendance duties	Equilibrium		2	2
duties, jugs	,Attendance duties	dustrial consideration: Reaction evaluation		2	3
duties, jugs	,Attendance duties	Types of industrial reactors		2	4
duties, jugs	,Attendance duties	nthesis Gas processes: gas production and steam reforming		2	5
duties, jugs	,Attendance duties	nthesis gas processes: ammonia: synthesis, oxidation		2	6
duties, jugs	,Attendance duties	ethanol synthesis and conversion		2	7
duties, jugs	,Attendance duties	Petroleum Refining: mposition, distillation, catalytic cracking, catalytic reforming, hydrotreating and coking		2	8
duties, jugs	,Attendance duties	Petroleum Refining: Alkylation and isomerization, steam cracking.		2	9
duties, jugs	,Attendance duties	Ethylene-Based Processes: Ethylene oxide and ethylene glycol, polyethylene, vinyl chloride and PVC.		2	10
,duties jugs	Attenda ,nce duties	Propylene-Based Processes: Acrylic acid and acrylonitrile, PP and Ziegler-Natta chemistry		2	11
duties, jugs	,Attendance duties	C4-Based Processes: Butadiene, Isobutylene.		2	12
duties, jugs	,Attendance duties	BTX Processes: Styrene and polystyrene, polyethylene terephthalate (PET), Phenol,		2	13

		adipic acid and nylon, phthalic anhydride			
duties, jugs	,Attendance duties	Topics in bioengineering and neuroscience: Principles, classifications		2	14
duties, jugs	,Attendance duties	Topics in bioengineering and neuroscience: Example: rules, neural networks (NN), genetics, building NN.		2	15

Course Evaluation .11	
Jugs, homework, college assignments, reports	
Learning and teaching resources .12	
H. A. Wittcoff , B. G. Reuben, and J. S. Plotkin, "Industrial Organic Chemistry". USA: A John Wiley & Sons, Inc., 3rd ed., 2013.	Required textbooks (methodology if (available
	Main References (Sources)
	Recommended supporting books and ,references (scientific journals (...reports
K. Weissermel and Dr. H.-J. Arpe, "Industrial Organic Chemistry." Verlag Chemie , 5th ed., 2010	Electronic references, websites

Course Description Form

Course name: Engineering Mechanics (Statics) .1
:Course code .2**MCT** 115
Semester/Year: First Semester - First Stage .3

Date this description was prepared: 2025 .4
.Available attendance forms: in-class attendance, online attendance .5
Number of study hours (total) / Number of units (total): 150 hours / 6 .6**ECTS**
:Name of the course supervisor (if more than one name is mentioned) .7
:Name: Asst . Prof. Dr. Sanaa Jaafar Yassin Email
Course objectives .8

The course aims to provide first-year students with basic knowledge of engineering mechanics. It studies everything related to forces and motion and related concepts such as equilibrium, force analysis in structures, centers of gravity, moments of inertia, friction, and body motion. The course aims to enable students to access engineering by understanding how to perform correct engineering analysis and how to deal with laws, equations, illustrative drawings, and other data and link the data together to reach the outputs and enable the student to be able to analyze, deduce, and conclude.	اهداف المادة الدراسية

The main strategy that will be adopted in delivering this module is to encourage students' participation in the exercises, while at the same time refining and expanding their critical thinking skills. This will be achieved through classes, interactive tutorials.	Strategy

Course environment .10

Evaluation method	Learning method	Name of unit or topic	Required learning outcomes	watches	week
duties, jugs	,Attendance duties	One General Principles 1.2 Fundamental Concepts 1.3 Units of Measurement 1.4 The International System of Units 1.5 Numerical Calculations 1.6 General Procedure for Analysis		3	1

duties, jugs	,Attendance duties	Two Force Vectors 2.1 Scalars and Vectors 2.2 Vector Operations		3	2
duties, jugs	,Attendance duties	2.3 Vector Addition of Forces		3	3
duties, jugs	,Attendance duties	2.4 Addition of a System of Coplanar Forces		3	4
duties, jugs	,Attendance duties	Equilibrium of a Particle 3.1 Condition for the Equilibrium of a Particle		3	5
duties, jugs	,Attendance duties	3.2 The Free-Body Diagram 3.3 Coplanar Force Systems		3	6
duties, jugs	,Attendance duties	Force System Resultants 4.1 Moment of a Force— Scalar Formulation 4.2 Moment of a Force—Vector Formulation		3	7
duties, jugs	,Attendance duties	4.3 Principle of Moments 4.4 Moment of a Force about a Specific Axis 4.5 Moment of a Couple		3	8
duties, jugs	,Attendance duties	Equilibrium of a Rigid Body 5.1 Conditions for Rigid-Body Equilibrium 5.2 Free-Body Diagrams		3	9
duties, jugs	,Attendance duties	5.3 Equations of Equilibrium 5.4 Two- and Three-Force Members		3	10
duties, jugs	,Attendance duties	5.5 Free-Body Diagrams 5.6 Equations of Equilibrium		3	11
duties, jugs	,Attendance duties	Friction		3	12

		6.1 Characteristics of Dry Friction			
duties, jugs	,Attendance duties	6.2 Problems Involving Dry Friction		3	13
duties, jugs	,Attendance duties	Center of Gravity and Centroid 7.1 Center of Gravity, Center of Mass, and the Centroid of a Body		3	14
duties, jugs	,Attendance duties	7.2 Composite Bodies		3	15

Course Evaluation .11	
Jugs, homework, college assignments, reports	
Learning and teaching resources .12	
fundamentals of engineering mechanics	Required textbooks (methodology if (available
	Main References (Sources)
	Recommended supporting books and ,references (scientific journals (...reports
Special requirements (including, for example, workshops, periodicals, software, and websites)	Electronic references, websites

Course Description Form

Course Name: Arabic Language 1 .1
:Course code .2**MCT** 116
Semester/Year: First Semester - First Stage .3
Date this description was prepared: 2025 .4

.Available attendance forms: in-class attendance, online attendance .5

Number of study hours (total) / Number of units (total): 50 hours / 2 .6**ECTS**

:Name of the course supervisor (if more than one name is mentioned) .7

:Name: M.M. Elaf Jalil Majeed Email

Course objectives .8

1. Working on the rise In the language Arabic And the governorate Ali Ha from error 2. To enable the student to understand the beauty of the Qur'anic style and its rhetorical miracles, so that he can be the ideal model in formulation. and expression Development Student spelling skills 3. Help the student learn the method The correct way to write and read numbers and write the hamza All kinds 4. Reviewing Arabic grammar rules to avoid .making grammatical mistakes 5. Development The student's skills in appreciating the beauty of literary expression and imitating beautiful texts .in writing 6. Helping the student to distinguish common linguistic errors and guiding him to correct them 7. Development Memorization and recitation skills	Course objectives

	Strategy

Course environment .10

Evaluation method	Learning method	Name of unit or topic	Required learning outcomes	watches	week
duties, jugs	,Attendance duties	Rhetoric lesson The importance of the Arabic language in general and its -importance for non specialist students		2	1
duties, jugs	,Attendance duties	,Parts of speech inflected and		2	2

		uninflected nouns and verbs			
duties, jugs	,Attendance duties	its appendix		2	3
duties, jugs	,Attendance duties	Types of plurals and what is related With it		2	4
duties, jugs	,Attendance duties	The Five Names		2	5
duties, jugs	,Attendance duties	Writing numbers		2	6
duties, jugs	,Attendance duties	- Hamza Primary Hamzat al-Qat` and Hamzat al-Wasl		2	7
duties, jugs	,Attendance duties	The middle and final hamza		2	8
duties, jugs	,Attendance duties	The Linguistic Miracle in the Holy Quran: Examples and Applications		2	9
duties, jugs	,Attendance duties	five verbs		2	10
duties, jugs	,Attendance duties	Common linguistic errors		2	11
duties, jugs	,Attendance duties	Writing the letters Dhad, Dad, Taa Marbuta , and Haa		2	12
duties, jugs	,Attendance duties	Expression lesson and examples of methods The most famous literary books		2	13
duties, jugs	,Attendance duties	A poem by an ancient poet and a modern poet		2	14
duties, jugs	,Attendance duties	discussion		2	15

Course Evaluation .11	
Jugs, homework, college assignments, reports	
Learning and teaching resources .12	
Clear Grammar / Ali Al-Jarim Rhetorical Touches / Fadhel Al-Samarrai -Arabic language curriculum for non specialized departments	Required textbooks (methodology if (available
	Main References (Sources)
	Recommended supporting books and ,references (scientific journals (...reports

/ linguistic errors Common Grammar Khader Abu Al-Ainain Meanings of Grammar / Fadhel Al-Samarrai Complete Poetic Works / Nazik Al-Malaika The Mu'allaqat /	Electronic references, websites

Course Description Form

Course Name: Computer Engineering Drawing .1	
:Course code .2**MCT** 117	
Semester/Year: First Semester - First Stage .3	
Date this description was prepared: 2025 .4	
.Available attendance forms: in-class attendance, online attendance .5	
Number of study hours (total) / Number of units (total): 100 hours / 4 .6**ECTS**	
:Name of the course supervisor (if more than one name is mentioned) .7	
:Name: M.M. Firas Matar Khalaf Email	
Course objectives .8	
1. Preparing and qualifying specialized engineers to meet the requirements of the labor market in the private and public sectors in mechanical engineering through diversifying the methods of learning, teaching and training students to apply the acquired knowledge and skills to solve real problems. 2. Providing distinguished academic programs in the field of mechanical engineering, both theoretical and practical, and international rules of academic quality that meet the needs of the labor market.	اهداف المادة الدراسية

3. Encouraging and developing scientific research in the fields of mechanical engineering. 4. Preparing a stimulating environment for faculty members to develop their knowledge and educational and research skills	
1. Explanation and clarification through lectures. 2. The method of displaying scientific materials on display devices: data show, smart boards, and plasma screens. 3. Self-learning through homework and mini-projects within the lectures. 4. Halls of application for engineering drawing 5. Graduation projects. 6. Summer training through the application in the design department	Strategy

Course environment .10

Evaluation method	Learning method	Name of unit or topic	Required learning outcomes	watches	week
duties, jugs	,Attendance duties	Engineering Drawing and Instruments +Drawing paper specification+ Duties		4	1
duties, jugs	,Attendance duties	Basic engineering operations+ duties		4	2
duties, jugs	,Attendance duties	Kinds of lines + Duties		4	3
duties, jugs	,Attendance duties	First angle projection + Duties		4	4
duties, jugs	,Attendance duties	Third angle projection + Duties		4	5
duties, jugs	,Attendance duties	Multi view projection + Duties		4	6
duties, jugs	,Attendance duties	Views distributions + Duties		4	7
duties, jugs	,Attendance duties	Drawing of side and top views + Duties		4	8
duties, jugs	,Attendance duties	Dimensioning of drawing+ Duties		4	9
duties, jugs	,Attendance duties	Full section + Duties		4	10
duties, jugs	,Attendance duties	Half section + Duties		4	11
duties, jugs	,Attendance duties	Isometric drawing + Duties		4	12
duties, jugs	,Attendance duties	Electrical drawing + Duties		4	13
duties, jugs	,Attendance duties	Electrical symbols + Duties		4	14
duties, jugs	,Attendance duties	Electronics symbols + Duties		4	15

Course Evaluation .11	
Jugs, homework, college assignments, reports	
Learning and teaching resources .12	
1- Textbook of Engineering Drawing Second Edition, K. Venkata Reddy	Required textbooks (methodology if (available
	Main References (Sources)
	Recommended supporting books and ,references (scientific journals (...reports
machine drawing, 3rd edition, Dr. K.L.Narayana , Dr. P.Kannaiah , and K.Venkata Reddy Electrical drawing, JC Cluley, 1990	Electronic references, websites

Course Description Form

Course Name: Mathematics 2 .1
:Course code .2**MCT** 121
Semester/Year: Second Semester - First Stage .3
Date this description was prepared: 2025 .4
.Available attendance forms: in-class attendance, online attendance .5

Number of study hours (total) / Number of units (total): 150 hours / 6 .6**ECTS**					
:Name of the course supervisor (if more than one name is mentioned) .7					
:Name: M.M. Youssef Abdel Wahab Khairallah Email					
Course objectives .8					
Learning Transcendental functions, Inverse trigonometric, Hyperbolic functions, their differentiation and integration, solving matrices, solving equations using arrays, and solving complex numbers.			Course objectives		
Attending lectures, viewing video lectures, solving homework assignments, and solving some questions from the prescribed book.				Strategy	
Course environment .10					
Evaluation method	Learning method	Name of unit or topic	Required learning outcomes	watches	week
duties, jugs	,Attendance duties	Transcendental functions (part 1)		3	1
duties, jugs	,Attendance duties	Transcendental functions (part 2)		3	2
duties, jugs	,Attendance duties	Hyperbolic functions		3	3
duties, jugs	,Attendance duties	Inverse trigonometric functions		3	4
duties, jugs	,Attendance duties	Method of integrations (part 1)		3	5
duties, jugs	,Attendance duties	Method of integrations (part 2)		3	6
duties, jugs	,Attendance duties	Method of integrations (part 3)		3	7
duties, jugs	,Attendance duties	Method of integrations (part 4)		3	8
duties, jugs	,Attendance duties	Matrices		3	9
duties, jugs	,Attendance duties	Determents		3	10
duties, jugs	,Attendance duties	Solve linear equations by matrices		3	11
duties, jugs	,Attendance duties	Complex numbers (part 1)		3	12

duties, jugs	,Attendance duties	Complex numbers (part 2)		3	13
duties, jugs	,Attendance duties	Polar coordinates (part 1)		3	14
duties, jugs	,Attendance duties	Polar coordinates (part 2)		3	15

Course Evaluation .11	
Jugs, homework, college assignments, reports	
Learning and teaching resources .12	
Thomas' Calculus Early Transcendentals	Required textbooks (methodology if (available
	Main References (Sources)
	Recommended supporting books and ,references (scientific journals (...reports
A Textbook of ENGINEERING MATHEMATICS-I	Electronic references, websites

Course Description Form

Course Name: Electrical Circuits 2 .1
:Course code .2**MCT** 122
Semester/Year: Second Semester - First Stage .3
Date this description was prepared: 2025 .4
.Available attendance forms: in-class attendance, online attendance .5

Number of study hours (total) / Number of units (total): 150 hours / 6 .6**ECTS**
:Name of the course supervisor (if more than one name is mentioned) .7
:Name: Asst . Prof. Dr. Alaa Falah Abdul Hassan Email
Course objectives .8

The theoretical foundations of computer engineering have expanded substantially in recent years. The objective of this course is to introduce students to this fundamental area of electrical circuits which enables students to focus on the study of electrical circuits. This course is designed to give the students an introduction to electrical currents, voltages, and the different elements of AC circuits. Also, it teaches them how to analyze AC circuits in steady and transient states.	اهداف المادة الدراسية
Explanation and clarification through lectures. - The method of displaying scientific materials electronically in the various programs within e-learning. - Self-learning through homework and direct questions within the lectures. - Material practical laboratories.	الاستراتيجية

Course environment .10

Evaluation method	Learning method	Name of unit or topic	Required learning outcomes	watches	week
duties, jugs	,Attendance duties	Transient response of an RC circuits		3	1
duties, jugs	,Attendance duties	AC quantities resistance, reactance, impedance, conductance, susceptance, and admittance, (Part 1)		3	2
duties, jugs	,Attendance duties	AC quantities resistance, reactance, impedance, conductance, susceptance, and admittance, (Part 2)		3	3
duties, jugs	,Attendance duties	peak values, maximum, average, and rms values, (Part1)		3	4
duties, jugs	,Attendance duties	peak values, maximum, average,		3	5

		and rms values, (Part 2)			
duties, jugs	,Attendance duties	phasor quantities. AC circuit analysis (equivalent impedance (Part 1))		3	6
duties, jugs	,Attendance duties	phasor quantities. AC circuit analysis (equivalent impedance (Part 2))		3	7
duties, jugs	,Attendance duties	power in AC circuit, power factor (Part 1)		3	8
duties, jugs	,Attendance duties	power in AC circuit, power factor (Part 2)		3	9
duties, jugs	,Attendance duties	Complex quantity, complex power, power factor correction		3	10
duties, jugs	,Attendance duties	Series resonance, quality factor, selectivity, half power frequencies and bandwidth.		3	11
duties, jugs	,Attendance duties	parallel resonance, quality factor, selectivity, half power frequencies and bandwidth.		3	12
duties, jugs	,Attendance duties	parallel resonance, quality factor, selectivity, half power frequencies and bandwidth.		3	13
duties, jugs	,Attendance duties	Single phase, and three phase circuits, star/delta transformation.		3	14
duties, jugs	,Attendance duties	Review		3	15
Course Evaluation .11					
Jugs, homework, college assignments, reports					
Learning and teaching resources .12					
Fundamentals of electric circuits, Charles K. Alexander | Matthew no Sadiku, 5th edition. 2- Introduction to circuit analysis, Boylested, 11th edition.			Required textbooks (methodology if (available		
			Main References (Sources)		

	Recommended supporting books and ,references (scientific journals (...reports
3- Engineering circuit analysis, William Hayte, 6th edition	Electronic references, websites

Course Description Form

Course Name: Digital Systems Design .1
:Course code .2**MCT** 123
Semester/Year: Second Semester - First Stage .3
Date this description was prepared: 2025 .4
.Available attendance forms: in-class attendance, online attendance .5
Number of study hours (total) / Number of units (total): 125 hours / 5 .6**ECTS**
:Name of the course supervisor (if more than one name is mentioned) .7
:Name: M.M. Rafid Mohammed Khalifa Email
Course objectives .8

This course aims to enable the student to learn basics of digital systems design: Numbering Systems, codes, and Conversion between different numbering systems, principles and laws of Boolean algebra, simplifying logical functions using k-map.	Course objectives
1. Provide lectures and printed sources from the modern, diverse and rich sources including examples. 2. Resolving some questions, with the intention of containing mistakes and making the students extract errors. 3. Asking questions, inquiries and solutions on the	Strategy

blackboard. 4. Questions directly for all students to learn the extent of interaction between the lecturer and students.	

Course environment .10					
Evaluation method	Learning method	Name of unit or topic	Required learning outcomes	watches	week
duties, jugs	,Attendance duties	Introduction to Digital Systems [Decimal, Binary, Octal, Hexadecimal, etc.] and Number – Base Conversions		3	1
duties, jugs	,Attendance duties	Arithmetic operations		3	2
duties, jugs	,Attendance duties	Complements of Numbers		3	3
duties, jugs	,Attendance duties	Binary Logic Gates and Discussion		3	4
duties, jugs	,Attendance duties	Basic Definition and Rules of Boolean Algebra		3	5
duties, jugs	,Attendance duties	Canonical and Standard Forms [sum of products, product of sums]		3	6
duties, jugs	,Attendance duties	Discussion		3	7
duties, jugs	,Attendance duties	The Karnough Map Method		3	8
duties, jugs	,Attendance duties	Don't-Care Terms		3	9
duties, jugs	,Attendance duties	NAND and NOR Implementation		3	10
duties, jugs	,Attendance duties	Logic Circuits		3	11
duties, jugs	,Attendance duties	Discussion		3	12
duties, jugs	,Attendance duties	Weighted Codes [BCD, etc.]		3	13
duties, jugs	,Attendance duties	Ex – n Codes and Gray code		3	14
duties, jugs	,Attendance duties	Design of different codes		3	15

Course Evaluation .11
Jugs, homework, college assignments, reports

Learning and teaching resources .12	
1- Digital Design, 4th Edition, by M. Morris Mano. Prentice-Hall, Inc. 2006 2- Digital	Required textbooks (methodology if (available

Computer Fundamental, by Thomas C. Bartee	
	Main References (Sources)
	Recommended supporting books and ,references (scientific journals (...reports
3 - Digital System Principle and Application, by Ronald J. Tocci	Electronic references, websites

Course Description Form

Course name: Physics .1	
:Course code .2**MCT** 124	
Semester/Year: Second Semester - First Stage .3	
Date this description was prepared: 2025 .4	
.Available attendance forms: in-class attendance, online attendance .5	
Number of study hours (total) / Number of units (total): 75 hours / 3 .6**ECTS**	
:Name of the course supervisor (if more than one name is mentioned) .7	
:Name: Asst . Prof. Dr. Muhannad Hamed Khalaf Email	
Course objectives .8	
This course aims to introduce students to the fundamental field of electronic elements and their applications.	Course objectives
1. Explanation and clarification using the lectures. 2. The methods of displaying the scientific materials using: data show, smart boards, plasma screens, and on-line meetings. 3. Self-learning using homework and small	Strategy

projects. 4. Laboratories. 5. Graduation projects. 6.Scientific visits. 7. Seminars held by the department. 8. Summer training.	

Course environment .10

Evaluation method	Learning method	Name of unit or topic	Required learning outcomes	watches	week
duties, jugs	,Attendance duties	Basics		2	1
duties, jugs	,Attendance duties	Basics		2	2
duties, jugs	,Attendance duties	Basics		2	3
duties, jugs	,Attendance duties	Basics		2	4
duties, jugs	,Attendance duties	Analyze movement of particles		2	5
duties, jugs	,Attendance duties	Analyze movement of particles		2	6
duties, jugs	,Attendance duties	Charge carriers in semiconductors		2	7
duties, jugs	,Attendance duties	Charge carriers in semiconductors		2	8
duties, jugs	,Attendance duties	Charge carriers in semiconductors		2	9
duties, jugs	,Attendance duties	Background for diodes		2	10
duties, jugs	,Attendance duties	Background for diodes		2	11
duties, jugs	,Attendance duties	Applications		2	12
duties, jugs	,Attendance duties	Applications		2	13
duties, jugs	,Attendance duties	Background for BJT		2	14
duties, jugs	,Attendance duties	Background for BJT		2	15

Course Evaluation .11

Jugs, homework, college assignments, reports

Learning and teaching resources .12

D. A. Neamen , Semiconductor Physics and Devices: Basic Principles. USA: McGraw-Hill, 4th ed., 2014. 2012	Required textbooks (methodology if (available
	Main References (Sources)
	Recommended supporting books and ,references (scientific journals (...reports

DA Neamen , Microelectronics: Circuit Analysis and Design. USA: McGraw-Hill, 4th ed., 2010.	Electronic references, websites

Course Description Form

Course name: Engineering Mechanics (Dynamics) .1	
:Course code .2**MCT** 125	
Semester/Year: Second Semester - First Stage .3	
Date this description was prepared: 2025 .4	
.Available attendance forms: in-class attendance, online attendance .5	
Number of study hours (total) / Number of units (total): 150 hours / 6 .6**ECTS**	
:Name of the course supervisor (if more than one name is mentioned) .7	
:Name: Asst . Prof. Dr. Sanaa Jaafar Yassin Email	
Course objectives .8	
The main objective of this course is to provide a basic understanding / concept, of principle, laws, and applications of Engineering Mechanics. The course will also discuss specific applications of Engineering Mechanics in environmental fields as well.	Course objectives
1. Explanation and clarification using the lectures. 2. The methods of displaying the scientific materials using: data show, smart boards, plasma screens, and on-line meetings. 3. Self-learning using homework and small projects. 4. Laboratories. 5. Graduation projects. 6.Scientific visits. 7. Seminars held by the department. 8. Summer training.	Strategy
Course environment .10	

Evaluation method	Learning method	Name of unit or topic	Required learning outcomes	watches	week
duties, jugs	,Attendance duties	Kinematics of Particles		3	1
duties, jugs	,Attendance duties	. Rectilinear Motion of Particles POSITION, VELOCITY, AND ACCELERATION		3	2
duties, jugs	,Attendance duties	Uniform Rectilinear Motion		3	3
duties, jugs	,Attendance duties	Motion of Several Particles		3	4
duties, jugs	,Attendance duties	Dependent Motions		3	5
duties, jugs	,Attendance duties	Curvilinear Motion of Partials (Position Vector, Velocity, Acceleration)		3	6
duties, jugs	,Attendance duties	Kinetics of Particles		3	7
duties, jugs	,Attendance duties	Newton's Second Law		3	8
duties, jugs	,Attendance duties	Newton's second law of motion		3	9
duties, jugs	,Attendance duties	System of Units		3	10
duties, jugs	,Attendance duties	Equation of motion A- Rectangular Components B-Tangential and Normal Components		3	11
duties, jugs	,Attendance duties	Energy and Momentum Methods		3	12
duties, jugs	,Attendance duties	WORK OF A FORCE		3	13
duties, jugs	,Attendance duties	Work of the Force of Gravity		3	14
duties, jugs	,Attendance duties	PRINCIPLE OF WORK AND ENERGY		3	15

Course Evaluation .11	
Jugs, homework, college assignments, reports	
Learning and teaching resources .12	
Engineering mechanics (Dynamics), by JL Mariam and LG Kraige	Required textbooks (methodology if (available
	Main References (Sources)

	Recommended supporting books and ,references (scientific journals (...reports
Theory and problems of engineering mechanics (statics and dynamics)	Electronic references, websites

Course Description Form

Course Name: English Language 1 .1	
:Course code .2**MCT** 126	
Semester/Year: Second Semester - First Stage .3	
Date this description was prepared: 2025 .4	
.Available attendance forms: in-class attendance, online attendance .5	
Number of study hours (total) / Number of units (total): 50 hours / 2 .6**ECTS**	
:Name of the course supervisor (if more than one name is mentioned) .7	
:Name: M.M. Mohammed Jassim Mohammed Radhi Email	
Course objectives .8	
1. To focus on practical programs to develop the skills of reading, listening, speaking, and writing in the English language. 2. To contribute to the preparation of various studies and research to qualify graduates in the English language. 3. To study English literature in its various specializations in different historical periods. 4. To develop the spirit of creativity and critical thinking in the learner, and to prepare them to criticize and analyze books and literary publications.	اهداف المادة الدراسية

1. Self-Directed Learning: Encouraging students to take responsibility for their own learning and seek out learning resources. 2. Collaborative Learning: Encouraging students to learn with their peers and share knowledge. 3. Project-Based Learning: Encouraging students to work on real-life engineering projects to learn English language skills. 4. E-Learning: Using technology to learn English language skills, such as e-learning materials and platforms. 5. Activity-Based Learning: Encouraging students to participate in interactive learning activities. Teaching Techniques 1. Audio-Visual Techniques: Using sound and image to teach English language skills. 2. Interactive Techniques: Using technology to teach English language skills, such as educational games and interactive activities. 3. Internet-Based Techniques: Using the internet to teach English language skills, such as e-learning materials and platforms. Learning Techniques 1. Computer-Based Techniques: Using computers to learn English language skills, such as educational software and interactive activities. 2. Internet-Based Techniques: Using the internet to learn English language skills, such as e-learning materials and platforms. 3. Mobile-Based Techniques: Using mobile devices to learn English language skills, such as educational apps and interactive activities.	Strategy

Course environment .10

Evaluation method	Learning method	Name of unit or topic	Required learning outcomes	watches	week
duties, jugs	,Attendance duties	Names (Part 1)		2	1
duties, jugs	,Attendance duties	Names (Part 2)		2	2
duties, jugs	,Attendance duties	Verbs (Part 1)		2	3
duties, jugs	,Attendance duties	Verbs (Part 2)		2	4
duties, jugs	,Attendance duties	Greetings (Part 1)		2	5
duties, jugs	,Attendance duties	Greetings (Part 2)		2	6
duties, jugs	,Attendance duties	Prepositions (Part 1)		2	7
duties, jugs	,Attendance duties	Prepositions (Part 2)		2	8
duties, jugs	,Attendance duties	Email Creating (Part 1)		2	9
duties, jugs	,Attendance duties	Email Creating (Part 2)		2	10

duties, jugs	,Attendance duties	Grammar (Part 1)		2	11
duties, jugs	,Attendance duties	Grammar (Part 2)		2	12
duties, jugs	,Attendance duties	Speaking & Writing		2	13
duties, jugs	,Attendance duties	Speaking & Writing		2	14
duties, jugs	,Attendance duties	Review		2	15

Course Evaluation .11	
Jugs, homework, college assignments, reports	
Learning and teaching resources .12	
Grammar in use Intermediate, 3rd edition, by Raymond Murphy	Required textbooks (methodology if (available
	Main References (Sources)
	Recommended supporting books and ,references (scientific journals (...reports
Grammar in use Intermediate, 3rd edition, by Raymond Murphy	Electronic references, websites

Course Description Form

Course name: Human Rights, Democracy and Freedom .1
:Course code .2**MCT** 127
Semester/Year: Second Semester - First Stage .3
Date this description was prepared: 2025 .4
.Available attendance forms: in-class attendance, online attendance .5

Number of study hours (total) / Number of units (total): 50 hours / 2 .6**ECTS**
:Name of the course supervisor (if more than one name is mentioned) .7
:Name: M.M. Ahmed Shatti Ghadhban Email
Course objectives .8

Human rights concepts, values and patterns of -:1 .thinking that lead to respect for human rights Ensure focus on national identity, respect it, and -2 .work to consolidate its principles Establishing and respecting human rights -:3 .principles and working to mature them Raising a generation that is aware and believes -4 in the values of tolerance, diversity and difference through respecting human dignity and activating .the mind and thinking And creativity	Course objectives
.Assigning the student to work on submitting the assigned tasks .Teamwork, supervising student groups and communicating with them 2-	Strategy

Course environment .10

Evaluation method	Learning method	Name of unit or topic	Required learning outcomes	watches	week
duties, jugs	,Attendance duties	The linguistic and technical meaning of the concept of human and right		2	1
duties, jugs	,Attendance duties	The basis of the idea of right and the most important theories that dealt with the definition of right 1- Voluntary theory 2- Objective theory		2	2
duties, jugs	,Attendance duties	3 - Mixed theory 4 - Modern theory Statement of the importance and characteristics of human rights		2	3
duties, jugs	,Attendance duties	The historical development of human rights in ancient legal systems 1- Primitive societies		2	4

		2 - Tribal societies			
duties, jugs	,Attendance duties	The historical development of human rights in ancient civil societies 1- 'Mesopotamias interest in human rights 2- The Pharaohs' interest in human rights		2	5
duties, jugs	,Attendance duties	Human rights in European civilization 1- Human rights in Greek civilization 2 - Human rights in Roman civilization 3 - Human rights in European civilization Middle Ages))		2	6
duties, jugs	,Attendance duties	Midterm exam		2	7
duties, jugs	,Attendance duties	Manifestations of Islamic Sharia's interest in human rights		2	8
duties, jugs	,Attendance duties	Human rights in international covenants and declarations		2	9
duties, jugs	,Attendance duties	Types of public rights and freedoms Personal rights and freedoms 1 - The right to life 2- The right to dignity 3 - The right to litigate 4 - Freedom of movement and residence 5 - The right to nationality Intellectual rights and freedoms The right to participate in the		2	10

		management of state affairs Economic freedoms and social rights			
duties, jugs	,Attendance duties	Intellectual rights and freedoms 1- Freedom of belief and religion 2- Freedom of opinion 3- Right of assembly 4- Freedom of the press 5- Freedom of ,radio, television cinema and theatre 6- freedom of education 7- The right to form associations .and political parties		2	11
duties, jugs	,Attendance duties	Rights and Freedoms in the Constitution of the Republic of Iraq of 2005 Intellectual rights and freedoms		2	12
duties, jugs	,Attendance duties	Intellectual rights and freedoms Economic freedoms and social rights		2	13
duties, jugs	,Attendance duties	Means of protecting human rights Legal means Judicial means political means		2	14
duties, jugs	,Attendance duties	Human rights in the face of new challenges terrorism human trafficking Administrative corruption		2	15
Course Evaluation .11					
Jugs, homework, college assignments, reports					
Learning and teaching resources .12					

, Ammar Abbas Al-Husseini, Human Rights 4th ed ., Dar Al-Salam Legal Library ., Cairo, 2022	Required textbooks (methodology if (available
	Main References (Sources)
	Recommended supporting books and ,references (scientific journals (...reports
, Ammar Abbas Al-Husseini, Human Rights 4th ed ., Dar Al-Salam Legal Library ., Cairo, 2022	Electronic references, websites